Assignment 9 — Network Security

Subject: Computer Networks
Assignment No.: 09
Title: Network Security and Common Cyber Threats
Difficulty: Intermediate
Submission Format: PDF / DOCX

Objective

To understand network security concepts and methods used to protect data and network resources.

Questions

1. What is network security?
2. Explain the CIA Triad.
3. Explain confidentiality.
4. Explain integrity.
5. Explain availability.
6. What is a firewall?
7. Explain encryption and decryption.
8. What is authentication?
9. What is authorization?
10. Explain access control.
11. What is antivirus software?
12. What is phishing?
13. Explain malware.
14. Explain Man-in-the-Middle attacks.
15. Explain Denial-of-Service attacks.
16. What is a digital signature?
17. What is a digital certificate?
18. Differentiate between digital signature and digital certificate.
19. Explain five methods for improving network security.
20. Explain how security can be implemented in StudySphere.